Abhishek Donekal

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PROFESSIONAL SUMMARY:

Data Engineer and MLOps specialist with 5 years of experience building and deploying AI solutions at scale. Expert in architecting data pipelines and multi-cloud infrastructure across AWS and GCP, with a focus on Generative AI, LLMs, and AI agents. Proven track record of optimizing big data processing and implementing CI/CD best practices to drive reliability and business value.

TECHNICAL SKILLS

- **Cloud, Infrastructure & Distributed Systems:** AWS (S3, EC2, Lambda, SageMaker, EMR, Redshift, CodePipeline, CloudWatch, EKS, Glue), Azure (Data Factory, Databricks, Synapse Analytics, AKS, Monitor, Functions, AI Foundry), GCP (BigQuery, Cloud Storage, Dataproc, GKE, Dataflow, Stackdriver), Oracle Cloud Infrastructure (OCI 2025); Apache Spark (Core, SQL, Streaming), PySpark, Spark SQL, Delta Lake, Hive, Pig, HDFS, MapReduce, YARN, Sqoop, Oozie.
- **Data Engineering, AI & Machine Learning:** PyTorch, TensorFlow, Keras, MLflow, LangChain, LlamaIndex, Auto-GPT, CrewAI, Haystack, Hugging Face Transformers, OpenAI/Anthropic/Gemini API Integration, LLM Orchestration, Distributed Training (Ray, GPU Clusters), GANs, VAEs, Diffusion Models, CUDA, cuDNN, Scikit-learn, NLTK, SpaCy, Gensim, OpenCV, YOLO; Vector Databases (FAISS, Pinecone, Qdrant, ChromaDB, Milvus); MLOps & Observability (Kubeflow, BentoML, Arize AI, Weights & Biases); Data Formats (JSON, Parquet, Avro, CSV).
- **Programming, Databases & Middleware:** Python, SQL, Scala, Go, R, Bash/Shell Scripting, PL/SQL; Databases (Snowflake, Amazon Redshift, Azure Synapse, MongoDB, Cassandra, MySQL, Oracle, PostgreSQL, Teradata, Elasticsearch); Messaging & Networking (Apache Kafka, Confluent, RTI DDS, Wireshark, SNMP); OS (Linux/Unix - RHEL & Ubuntu, Windows).
- **DevOps, Automation & Governance:** Terraform (IaC), Ansible, Kubernetes, Docker, Helm, Jenkins, Git, GitHub Actions, Azure DevOps, Bitbucket, JFrog Artifactory, GCR; Monitoring & Observability (Prometheus, Grafana, ELK Stack, Zabbix, CloudWatch); Governance & Security (Lake Formation, Glue Data Catalog, KMS, HIPAA, PCI DSS, Kerberos, Apache Ranger, RBAC); Project Methodologies (JIRA, Agile, ServiceNow).

Certifications, Knowledge Badges and Digital Portfolio :

- AWS Sysops administrator associate - Credly link
- Google Cloud Professional Devops Engineer Credly Link
- Google Associate Cloud Engineer Certification Credly link
- Databricks Generative AI Fundamentals
- LLaMA 2 Fine-Tuning: Automated Resume Customizer Github link
- Microsoft Certified: Azure Cosmos Db Develop
- Microsoft Certified: Azure Data Scientist Associate
- AWS & Infrastructure Automation (Terraform/Streamlit) Github Link
- RAG-based PDF Chatbot Github Link
- LangChain Chat with your Data Cloudera link

Education :

Florida International University, **MS in Data Science,** Spring 2024
Amrita Vishwa vidhyapeetham, **Bachelor's of Technology**, Fall 2020

WORK EXPERIENCE:
Carnival Cruises corporate | Miami, Fl December 2024 to Present
Data Engineer

- Architected and maintained scalable Big Data systems using Agile methodologies, integrating Hadoop HDFS with HiveDB to improve enterprise data processing reliability.
- Optimized MapReduce and Apache Spark jobs through strategic data partitioning, **reducing job latency by 35%** through performance tuning and resource optimization.
- Automated SASWORK cleanup using Python and SQL scripts, enhancing cluster stability and ensuring seamless performance during high-intensity Agile sprints.
- Engineered direct SAS-to-Hadoop connections via SQL Pass-Through to HDFS, accelerating data access by eliminating manual download steps from the pipeline.
- Boosted data throughput by tuning connection parameters and execution plans for **multi-terabyte datasets**, resulting in significantly faster reporting cycles.
- Led iterative development cycles within a Scrum framework, leveraging Jira to consistently meet stakeholder requirements through efficient data engineering task management.
- Orchestrated complex data pipelines and automated cluster health monitoring using Excel and VBA, maintaining a consistent **99.9% system uptime**.
- Developed HiveQL schemas and Excel Power Query models that **improved query performance by 40%** using Vectorization and Cost-Based Optimization.
- Configured YARN resource allocation to support high-concurrency production environments, optimizing scheduler performance for stable resource management.
- Implemented Apache ZooKeeper for configuration management, ensuring high-availability for NameNodes and persistent SQL sessions during failovers.
- Orchestrated cluster upgrades and patch management using Agile release protocols, achieving **zero data loss** across production and staging environments.
- Secured the data lake with Kerberos and Apache Ranger, enforcing Role-Based Access Control (RBAC) and ensuring strict privacy compliance via SQL-based auditing.
- Created custom Shell and Python scripts to automate SQL health checks and HDFS Balancer tasks, maintaining optimal data distribution across nodes.
- Established a Hybrid Data Lake by integrating Hadoop with Cloud Storage and Excel, enabling cost-effective archival and access to historical data.
- Streamlined disaster recovery by configuring DistCp for data replication, helping BI teams meet their Recovery Time Objectives (RTO) for critical reporting.

Environment: Hadoop (HDFS, YARN), Apache Spark, HiveDB, Python, SQL, SAS/ACCESS, Jira, Kerberos, Apache Ranger, Cloud Storage (S3/GCS), Excel.

Mcdonald's corporate | Miami, Fl December 2024 to Present
Devops Engineer

- Spearheaded the end-to-end cloud migration of all critical ETL workflows from AWS Redshift to GCP BigQuery, resulting in **40% improved query performance** and a **30% reduction in operational costs**.
- Decreased service downtime by **50%** for high-availability AI services by migrating containerized deployments from AWS EKS to **GCP GKE**.
- Migrated and optimized Apache Spark pipelines from AWS EMR to **GCP Dataproc**, **decreasing processing time by 6x** and significantly improving cost efficiency in the new cloud environment.
- Lowered operational overhead by **55%** by re-platforming serverless applications from AWS Lambda to resilient GCP Cloud Functions.
- Cut infrastructure provisioning time by **70%** across the new GCP environment by implementing IaC using Terraform and GCP Deployment Manager, standardizing configuration post-migration.
- Engineered robust CI/CD pipelines to support the new MLOps architecture in GCP, reducing manual deployment intervention by **60%** and ensuring continuous delivery of models.
- Reduced system downtime by **45%** by establishing automated monitoring across GCP services (Stackdriver/Prometheus), which also tracked LLM performance metrics post-migration.
- Designed and implemented a unified data lake architecture with Delta Lake in GCP, achieving **2.8x faster data retrieval** for subsequent vector database integrations.

- Developed and fine-tuned LLMs (Llama 2) for enterprise AI solutions using Hugging Face Transformers within the scaled GCP infrastructure.

Environment: AWS (S3, EC2, Step Functions, Lambda, MWAA, Glue, API Gateway, CloudWatch), GCP (BigQuery, Cloud Storage, Dataproc, GKE, Dataflow, Stackdriver), Snowflake, Tableau, Terraform, Jenkins, Kafka, Teradata, Python, SQL

Energy systems research laboratory | Miami, Fl January 2023 - June 2024
MLOps Engineer
Responsibilities:

- Engineered and deployed scalable data pipelines in AWS, leveraging Lambda, S3, and EC2 for high-efficiency data management and transformation.
- Orchestrated AWS-based ETL workflows, transforming raw telemetry into strategic business assets and leveraging cloud infrastructure for automated model serving and scalable inference pipelines.
- Architected a Time-series database (MongoDB) to capture high-velocity data points using RTI Data Distribution Service (DDS) for a smart grid testbed.
- Scaled model training to multi-node GPU clusters, **decreasing experiment runtime by 8x** through parallel processing optimization.
- Architected and managed high-performance GPU clusters to support distributed training and large-scale model deployments.
- Configured and managed complex IP networking, including subnets, VLANS, NAT, and DHCP, while troubleshooting connectivity issues to optimize network performance.
- Resolved critical bottlenecks in the RTI DDS communication layer by analyzing packet captures with Wireshark to maintain low-latency data streams.
- Implemented automated CI/CD pipelines using GitHub Actions and AWS CodePipeline for the seamless testing, versioning, and deployment of ML models.
- Established model monitoring and observability using Prometheus and Grafana, tracking model drift and performance degradation in real-time.
- Containerized machine learning workflows with Docker and orchestrated deployments using Kubernetes (EKS) to ensure environment parity.
- Engineered end-to-end MLOps lifecycles using MLflow, Docker, and Kubernetes to ensure seamless production handoffs.
- Automated server health monitoring via Python and SNMP, providing proactive alerts for hardware failures within the high-performance GPU cluster.
- Engineered a centralized logging system using the ELK Stack (Elasticsearch, Logstash, Kibana) to aggregate system logs for rapid troubleshooting.

Environment: AWS (Lambda, S3, EC2, AWS CodePipeline, EKS), MongoDB, RTI DDS, Wireshark, GitHub Actions, Prometheus, Grafana, Docker, Kubernetes (EKS), MLflow, Python, SNMP, ELK Stack.

Eastman Chemicals pvt LTD | Hyderabad, IN January 2021 - June 2022
Automation Analyst
Responsibilities:

- Developed and optimized robust web-scraping pipelines using Python, BeautifulSoup, and Selenium to automate the collection of large-scale, unstructured data from dynamic web sources.
- Designed and deployed interactive, high-impact health and business dashboards in Tableau and Power BI, translating complex telemetry into automated monitoring dashboards to track system health and performance KPIs.
- Automated data extraction pipelines using Python, significantly reducing manual overhead and increasing overall system throughput.
- Automated data extraction and cleansing workflows to eliminate manual entry, significantly increasing data accuracy and reducing processing latency for downstream analysis.
- Integrated disparate data sources ranging from web-scraped APIs to SQL databases into a centralized BI framework to provide a unified view of performance metrics.
- Applied advanced data modeling techniques to identify trends and anomalies, effectively bridging the gap between raw technical data and strategic business requirements.
- Maintained and updated dynamic reporting suites, ensuring data integrity and real-time visualization of key performance indicators (KPIs) through optimized DAX and Power Query logic.

Environment: Python, BeautifulSoup, Selenium, Tableau, Power BI, SQL, DAX, Power Query.